

Industry Application of Frugal AI to Tabular Data Extraction from PDFs

A Cloud-to-Local Cascade Framework for Environmental Product Declarations

Interim Report

MSc Computing — Imperial College London

Author: Edward Xu

Supervisor: Dr Thomas Heinis

Industrial Partner: Firstplanit

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Chapter 1 — Introduction

1.1 The Problem

Business data is overwhelmingly locked inside PDF documents, and a large fraction of that data sits inside tables. Across finance, science, healthcare, and the construction industry, extracting that tabular data into structured form remains as a stubborn operational bottleneck in modern enterprise computing. In December 2025, Erich Elsen, Principal Research Scientist at Databricks, observed that despite the widespread assumption that PDF parsing is a solved problem, enterprise documents in production routinely defeat available tools through their combination of digital-native content, scanned pages, irregular layouts, and embedded tables [9]. Recent benchmarking confirms the gap is real but uneven: a 2026 evaluation of 21 contemporary PDF parsers across 451 tables found significant performance disparities, with parser choice mattering as much as the underlying document quality [10]. Performance varies markedly with table complexity, document type, and evaluation metric.

The emergence of large language models (LLMs) with strong general capabilities has accelerated what is possible in this space. Frontier cloud models can extract complex schemas from heterogeneous documents at a quality unreachable by previous rule-based or statistical methods, but this has produced its own problem. The dominant industry response has been to apply progressively larger general-purpose models to every document, regardless of complexity; defaulting to the strongest available model for every task, and in some cases treating high token consumption as itself a marker of successful adoption. By mid-2026 the economic consequences of this default had begun to draw sustained attention in the financial press. Reports from Fortune and Tom's Hardware in May 2026 documented internal practices at Microsoft, Meta, Amazon and Uber that explicitly encourage maximum token consumption, including internal leaderboards tracking employee AI usage [11,12]. Goldman Sachs has forecast a 24-fold increase in token consumption by 2030, driven primarily by agentic systems that can consume up to three orders of magnitude more tokens per task than conventional LLM interactions [11,12]. Concurrent reporting indicates that frontier providers have begun raising prices, imposing usage limits, and reducing default model capabilities - adjustments consistent with the unwinding of subsidised adoption pricing [13]. The viability of frontier-only deployment as a default architectural choice is, on the current trajectory, an open question.

This is the broader context in which the present project sits. The specific question it addresses is narrower: whether, for a representative high-volume industrial extraction problem, a more selective and frugal use of LLM capacity can achieve comparable extraction quality at materially lower cost, energy, and dependency on external providers.

1.2 Frugal AI and Local Models

A recent counter-movement, broadly grouped under the banner of *Frugal AI*, has formalised the argument against this default. The Frugal AI Hub at Cambridge Judge Business School, launched in 2025, defines frugality not as cost-cutting under constraint but as a deliberate design philosophy: matching the size and capability of the model to the difficulty of the task, and reserving frontier models for cases that genuinely require them [3,4]. The Hub identifies four pillars: **Resource Efficiency**,

Sustainability, Accessibility and Inclusion, and Impact and Scalability, and claims that adopting them can deliver 70–90% reductions in total cost of ownership and up to 80% reductions in energy use without sacrificing utility [3].

Parallel academic work supports this argument empirically. Chen, Zaharia and Zou's FrugalGPT [5] demonstrated on public NLP benchmarks that strategic cascading between models of varying cost can match GPT-4 performance at up to 98% lower cost. Ong et al.'s RouteLLM [6] showed over 2× cost reduction at maintained quality. Belcak et al. at NVIDIA argued in a recent position paper that small language models could be the future of agentic AI, reframing the SLM-versus-LLM debate from a capability question to a deployment question [7]. If intelligence is heading toward becoming a billed utility, like water or electricity, then frugal use of it is not optional in the long run.

A second development reinforces this argument. Open-weight small language models in the 3–14 billion parameter range, including Microsoft's Phi-4-Mini, Alibaba's Qwen 2.5 family, and Google's Gemma 4 [14,15], have become sufficiently capable to run on consumer-grade GPUs at near-zero marginal cost per inference. Parameter-efficient fine-tuning methods like LoRA and QLoRA [16,17] further permit adaptation to specific domains with fewer than 100 training examples on a single desktop GPU [18]. Whereas FrugalGPT cascaded between cheap and expensive cloud APIs, the current generation of local models admits an alternative configuration: cascading from a locally hosted, domain-fine-tuned SLM to a frontier model only when the input warrants it. This alters the cost, governance, and vendor-independence calculus in ways the original FrugalGPT framework did not anticipate.

1.3 What This Project Does

The Frugal AI and LLM–SLM collaboration literatures have, between them, established three components of an answer. FrugalGPT [5] showed on public NLP benchmarks that strategic cascading between models of varying cost can match frontier accuracy at 1–2% of the cost, formalising prompt adaptation, LLM approximation, and LLM cascade as a family of complementary strategies. RouteLLM [6] extended cascading into a learned routing problem. Liu, Lian et al.'s Strata study [18], published in Scientific Reports in late 2025, demonstrated that in a real institutional setting of clinical information extraction across four pathology and radiology datasets at UCSF, a LoRA-fine-tuned local 8-billion-parameter Llama model achieves non-inferior performance to the second human annotator on all four datasets tested, with average exact-match accuracy of 90.0 ± 1.7 using fewer than 100 training reports on a single desktop-grade GPU. The Frugal AI literature has argued for a path in which small, locally hosted, domain-fine-tuned models substitute for expensive cloud LLMs; Strata offers concrete evidence that this approach can work in a deployed pipeline.

What is still missing across this literature is a *combined* treatment. FrugalGPT evaluated cascading strategies on benchmarks without practical deployment constraints. Strata evaluated a single strategy of local fine-tuning in a single domain (clinical), without the cascading and prompt-adaptation components of the broader FrugalGPT framework. RouteLLM, the EMNLP 2025 SATER paper on routing-and-cascading [19], and the recent multi-agent financial-document benchmarking work [20] each evaluate one or two strategies in isolation. No published study evaluates the *full FrugalGPT triad*

combined in a deployed industrial pipeline, with the cascade modernised to begin from a domain-fine-tuned local SLM as Strata advocates rather than from a cheaper cloud API as FrugalGPT originally specified.

This project addresses that gap through collaboration with Firstplanit, a UCL-incubated sustainability data platform for the built environment [21,22]. Central to Firstplanit's operation is a scalable ingestion pipeline that continuously absorbs Environmental Product Declarations (EPDs), third-party-verified documents that report a construction product's environmental performance across its full life cycle, from a heterogeneous and constantly updating set of sources. The current pipeline relies on a single frontier cloud API (Gemini 2.5 Flash) and therefore carries the cost-scaling, vendor lock-in, and data-governance risks that the Frugal AI literature identifies as the motivation for frugal alternatives. The EPD extraction problem at Firstplanit is not a contrived case study but a live commercial bottleneck in which the proposed approach can deliver measurable industrial value.

Methodologically, the project takes the FrugalGPT framework as its organising spine and Strata as its operational template. It evaluates all three FrugalGPT strategies (prompt adaptation, LLM approximation, LLM cascade) both individually and in combination. Following Strata's empirical finding that <100 training examples and a single consumer GPU are sufficient for human-level performance, it applies QLoRA fine-tuning to a small local SLM (Phi-4-Mini, 3.8B) as the cascade's first stage. The cascade then escalates to mid-tier and frontier cloud models only when the local SLM's confidence is insufficient. Evaluation is benchmarked against Firstplanit's current production pipeline, against a frontier cloud reference, and against single-model local baselines, using metrics that go beyond accuracy to include cost per document, latency, energy use, schema validity, and confidence calibration.

1.4 What this Project Looks to Contribute

First, a domain extension. This is, to the best of the author's knowledge, the first published evaluation of frugal extraction strategies applied to EPDs. The Strata work demonstrated the local-fine-tuned-SLM approach in a clinical setting; this project tests whether the same approach generalises to a structurally different industrial extraction problem, such as commercial documents in a startup rather than a research hospital, EN 15804-governed sustainability data rather than pathology, a multi-programme-operator corpus rather than two academic medical centres. The findings are also however intended to be transferable: the project's failure-mode taxonomy and methodology generalise to the broader class of semi-structured tabular PDF extraction problems in finance, science, and adjacent domains.

Second, a methodological synthesis. The project takes the local-fine-tuned-SLM approach validated by Strata and combines it with the cascading and prompt-adaptation strategies of FrugalGPT, evaluated as a single integrated pipeline. Each component has been studied in isolation; the combination has not been studied in a deployed industrial setting. The combinatorial breadth of the evaluation of three strategies, multiple model sizes, individual and combined ablations is itself part of the contribution, providing the first systematic picture of how these frugal techniques interact when stacked rather than studied separately.

The project hypothesises that combining current proposals such as local fine-tuning and frugal cascading in a single deployed pipeline, applied to a real industrial problem involving messy and heterogeneous documents, produces an operating point on the cost-quality frontier that the prevailing frontier-only default cannot match on its own terms. The finding would have implications beyond EPDs for any organisation deploying LLMs at scale on tabular PDF data. Conversely, a significant drop in performance relative to prior studies would raise substantive questions about the applicability of frugal AI approaches and the actual capabilities of local LLMs in industrial settings.

While this remains an academic project first, the collaboration with Firstplanit also creates an opportunity for a direct industrial contribution: an optimised extraction pipeline that reduces inference cost and dependency on a single frontier provider while holding performance at or above the current production baseline. Findings are intended to be actionable for the partner, with the live commercial context serving as both a source of representative data and a benchmark against which the frugal approach can be meaningfully tested.

Chapter 2 — Background

This chapter establishes the technical and domain context that the literature reviews this project engages with. Section 2.1 introduces the PDF format and the structural reasons that tabular extraction is difficult. Section 2.2 introduces EPDs and the EN 15804 schema that defines their content. Section 2.3 outlines the language model landscape relevant to this work.

2.1 The PDF Format and the Extraction Problem

The Portable Document Format is standardised as ISO 32000 (currently version 2.0, 2020) [23], but it is fundamentally a document description format designed for visual fidelity rather than structured data interchange. A PDF describes *where to draw glyphs and lines on a page*, not what those glyphs *mean* as content. Two distinctions matter for extraction:

The first is the **tagged versus untagged** distinction. Tagged PDFs, typically those compliant with the PDF/UA accessibility standard (ISO 14289) [24], embed a logical structure tree that describes reading order, table boundaries, and semantic roles. Untagged PDFs carry no such structure and require inference from spatial position. The project currently speculates that an overwhelming majority of EPDs are untagged, which is subject to a more systematic audit.

The second is the **native versus scanned** distinction. Native PDFs contain searchable text streams produced directly from a source application; scanned PDFs are images of paper documents, requiring optical character recognition (OCR) before any textual processing. Older EPDs are disproportionately scanned, and standard text extraction libraries such as PyMuPDF fail silently on scanned input — returning empty strings without raising errors [25]. This silent failure mode is a significant operational risk in any production pipeline.

Within these structural categories, *tables* are one of the most problematic element for automated extraction. In EPDs the 168 indicator × life-cycle-stage values that form the analytic core of the document live in wide tables that defeat standard extractors: multi-row headers collapse, merged cells duplicate or fragment, column boundaries shift between pages, scientific notation splits across tokens, and negative-sign characters are encoded inconsistently (Unicode minus, hyphen-minus, or parenthetical notation). Dual-language European EPDs interleave glyph streams line by line, and multi-product PDFs share tables across stock-keeping units, requiring the extractor to isolate one column without contamination from neighbours.

These failure modes are not uniform across the corpus. A clean, single-product, tagged PDF from a modern programme operator demands far less of an extractor than a scanned, multi-product, dual-language EPD from a regional operator. This non-uniformity is the empirical foundation for the routing thesis developed in Chapter 3.

2.2 Environmental Product Declaration (EPD)

An Environmental Product Declaration is a standardised, third-party-verified document reporting the environmental performance of a construction product across its full life cycle. It functions as the built environment's nutrition label. EPDs are governed by ISO 14025 (Type III environmental declarations)

[26], EN 15804 (Product Category Rules for construction products) [27], and ISO 21930 [28], and are typically valid for five years from issue.

The reportable content of an EPD is highly structured but not fully standardised. The core analytical payload follows the EN 15804 indicator matrix: a set of environmental impact indicators (Global Warming Potential, Acidification Potential, Eutrophication Potential, Photochemical Ozone Creation Potential, Abiotic Depletion, and others) reported across life-cycle stages from raw material supply (A1) through end-of-life (C1–C4) and beyond (D). For this project, the extraction schema agreed with Firstplanit captures approximately 210 fields per document, of which approximately 168 form a structured indicator $\times$ stage matrix in which *N/A* is a valid response — a property with significant implications for evaluation, discussed in Chapter 3.

EPDs are issued by *programme operators*: organisations accredited to verify and publish declarations against the relevant Product Category Rules. Major operators include IBU (Germany), EPD International (Sweden), BRE (UK), EPD Hub, EPDItaly, PEP Ecopassport (France), and EPD Australasia. Each operator uses its own template, with consequences for layout, language, and table structure. There is no single global registry; aggregators such as EC3 / Building Transparency [29] provide centralised access to digitised EPDs, but the installed base of source documents is overwhelmingly PDF.

The openEPD standard [30] and the ECO Portal datahub [31] represent ongoing efforts to make EPDs natively machine-readable. EN ISO 22057 [32] further provides BIM-compatible data templates. These standards are emerging rather than ubiquitous, and the existing corpus of PDF declarations remains extraction-relevant for the foreseeable future. The EU Construction Products Regulation begins mandating EPD data in Declarations of Performance from mid-2026, with Digital Product Passport integration scheduled for 2028, creating a sharp near-term increase in extraction volume. This of course poses question to the relevance of this project and more broadly moat of Firstplanit, however in the short term this is unlikely to change, and this project seeks to explore more generally frugal pdf tabular data extraction and uses EPDs as a good data source which is underexplored.

2.3 Language Models, SLMs, and Guided Decoding

The language models relevant to this project fall on a spectrum of size, capability, and deployability. The conventional distinction between *large language models* and *small language models* is operational rather than principled: there is no agreed parameter threshold, but the Frugal AI Hub white paper and Belcak et al.'s NVIDIA position paper converge on roughly 10 billion parameters as the working boundary as of 2025 [3,7]. For this project, three tiers are relevant:

- **Small tier ($\leq 4\text{B}$ parameters):** Phi-4-Mini, Gemma 4 4B, Qwen 2.5 3B. Deployable on consumer GPUs; suitable for high-throughput, low-complexity documents.
- **Mid tier (7–14B):** Qwen 2.5 7B, Gemma 4 12B, Phi-4. Requires modest server hardware; viable as a local fallback tier.
- **Large open-weight (27B+):** Qwen 2.5 32B, GLM-4.5V, Qwen2.5-VL 72B. Approaches frontier capability at the cost of significant local GPU footprint.

A critical sub-category for this project is the **vision-language model** (VLM). Standard LLMs operate on tokenised text and require an upstream PDF-to-text step that frequently fails on tables. VLMs such as Qwen2.5-VL [33] consume the page image directly and reason jointly over visual layout and text content, bypassing the OCR/parsing bottleneck for table-heavy documents. The 3B and 7B Qwen2.5-VL variants are deployable on consumer hardware.

Guided decoding (also called constrained or structured decoding) is the technique where a language model is forced to produce output that conforms to a specified schema. At each generation step, tokens that would violate the target grammar are masked from the model's output distribution. Modern implementations include Outlines [34], XGrammar [35], and LlamaIndex's structured-output APIs. The relevance for this project is that guided decoding *guarantees* schema validity at the syntactic level — every output is parseable JSON conforming to the Firstplanit schema — but cannot guarantee semantic correctness. A model can still output structurally valid but factually wrong values. Guided decoding therefore reduces but does not eliminate the need for downstream validation.

Parameter-efficient fine-tuning is the third technique relevant to the project. Where the base capabilities of an SLM are insufficient for a task, full fine-tuning is prohibitively expensive. LoRA (Low-Rank Adaptation) [16] and its quantised variant QLoRA [17] reduce the trainable parameter count by orders of magnitude by inserting low-rank update matrices into a frozen base model, which enables fine-tuning of multi-billion-parameter models on a single consumer GPU. This is the mechanism by which the Strata work cited in Chapter 1 achieved human-level accuracy on desktop hardware [18], and represents a core methodological component of this project (Section 4.1.2).

Chapter 3 — Literature Review

This chapter is organised around the project's central thesis. Section 3.1 reviews the cost-quality routing literature centred on FrugalGPT that motivates the architectural choice. Section 3.2 surveys analogous extraction work in adjacent industrial domains, establishing that the paradigm proposed here is *already validated* outside the EPD context. Section 3.3 reviews the small but growing literature on EPD-specific extraction, identifying the precise gap this project fills. Section 3.4 covers supporting technical literature on structured extraction, layout-aware document understanding, and confidence calibration.

3.1 Cost-Quality Routing and Frugal Strategies

- **Prompt adaptation:** reducing the token cost per query by identifying shorter or more efficient prompts, for example through prompt selection (fewer in-context examples) or query concatenation (batching similar queries to amortise the prompt).
- **LLM approximation:** approximating an expensive model with cheaper alternatives on specific tasks, either by caching prior responses for reuse on similar queries or by fine-tuning a smaller model to imitate the expensive one's outputs.
- **LLM cascade:** querying models of increasing cost and capability in sequence, accepting the first sufficiently confident answer rather than always invoking the strongest model.

The empirical evaluation of FrugalGPT showed that a learned cascade could match GPT-4 performance at up to 98% lower cost on public NLP benchmarks (HEADLINES, OVERRULING, COQA, AGNEWS), establishing the cost-quality Pareto frontier as the relevant evaluation framework for frugal deployment rather than accuracy alone.

Subsequent work has extended each of the three strategies individually. On *cascade*, Ong et al.'s RouteLLM [6] reframed cascading as a learned routing problem trained on human preference data, demonstrating over 2× cost reduction at maintained quality. Shen et al.'s SATER, published at EMNLP 2025 [19], introduced refusal-trained SLMs that can themselves recognise when to escalate, reducing computational costs by over 50% as well as cascade latency by over 80%. On *LLM approximation*, the clinical extraction literature provides the most rigorous published evidence: in the Strata work [18], a LoRA-fine-tuned Llama-3.1 8B achieves non-inferior performance to a second human annotator on pathology and radiology reports using fewer than 100 training examples on desktop-grade hardware. On *prompt adaptation*, recent work on schema-aware compression and structured-output prompting [36,37] has continued to advance the state of the art.

The Cambridge Judge Frugal AI white papers [3,4] and Belcak et al.'s NVIDIA position paper [7] provide the broader philosophical and architectural framing under which these strategies sit. The Lu et al. survey of collaborative LLM-SLM mechanisms [38] organises the broader design space into pipeline collaboration, parallel collaboration, conditional triggered inference, and knowledge distillation, while Varangot-Reille et al. [39] survey routing strategies specifically.

Two gaps in this literature motivate the present project.

First, almost all published evaluations have been conducted on synthetic benchmark datasets. FrugalGPT itself was evaluated on public NLP corpora; RouteLLM on MT-Bench, MMLU, and GSM8K; SATER on six standard benchmarks. The financial-document benchmarking work of Kulkarni and Kulkarni [20] is the closest analogue to a deployed evaluation but examines multi-agent orchestration patterns rather than the FrugalGPT triad specifically. Strata [18] is the exception which was subject to deployed evaluation, but is limited to a single strategy and datatype.

Second, A more systematic evaluation combining strategies proposed by multiple studies across frugal AI and local models, with deployment considerations. There is no published study, to the best of the author's knowledge, that evaluates *all three* FrugalGPT strategies in combination which replaces the original cheaper-cloud-API first stage with a domain-fine-tuned local SLM. This is the central methodological extension proposed by this project.

3.2 Validation in Adjacent Industrial Domains

A critical question for any methodological proposal is whether it has been validated outside the immediate problem. For frugal extraction strategies in heterogeneous industrial corpora, the answer is *yes*, across multiple industrial domains.

In **financial document extraction**, Kulkarni and Kulkarni's 2026 paper *Benchmarking Multi-Agent LLM Architectures for Financial Document Processing* [20] evaluates four architectures across five frontier and open-weight LLMs on 10,000 SEC filings consisting of 10-K, 10-Q, and 8-K forms. They were measured along five axes: field-level F1, document-level accuracy, end-to-end latency, cost per document, and token efficiency. The key finding from this study are as follows and directly motivates this project: reflexive architectures achieve the highest field-level F1 (0.943) however at 2.3× the cost of sequential baselines, while hierarchical architectures occupy the most favourable position on the cost-accuracy Pareto frontier (F1 0.921 at 1.4× cost). Hybrid configurations recovered 89% of the reflexive accuracy gains at only 1.15× baseline cost. This is the evaluation methodology and reporting structure that this project will look to adopt and extend.

In **invoice and form extraction**, Wang and Shen's 2025 paper *Hybrid OCR-LLM Framework for Enterprise-Scale Document Information Extraction Under Copy-heavy Task* [40] evaluates 25 configurations across three extraction paradigms on identity documents in four formats, achieving F1=1.0 accuracy at sub-second latency for structured documents. The paper's framing of "structure-aware method selection" as the route from computational burden to optimisation opportunity is a direct rhetorical and methodological parallel to the present work.

In **scientific and materials extraction**, the Schema-Driven Information Extraction work of Bai et al. [41] demonstrated the cross-domain transferability of schema-grounded extraction across machine learning papers, chemistry, materials science, and webpages.

In **clinical informatics**, the *Strata* work [18] is treated in this project as a co-equal precedent rather than mere background, and is discussed in detail in Chapter 1.3. It showed that LoRA-fine-tuned Llama-3.1 8B achieves non-inferior performance to a second human annotator across four datasets, using

fewer than 100 training reports and desktop-grade hardware. This is the strongest available evidence that the QLoRA component of this project is empirically tractable.

Across these domains, the same architectural pattern recurs: heterogeneous source documents, schema-grounded extraction, cost-quality benchmarking, and local-or-routed deployment to satisfy governance constraints. The present project is best understood as the extension of this validated paradigm to a previously under-studied document class of EPDs that shares all the relevant structural properties, alongside extension in the exploration of fine-tuned local LLMs.

3.3 EPD-Specific Extraction Literature

The published literature on automated EPD extraction is thin, which is itself a key motivation for this project.

The most directly relevant prior work is the 2025 NIST Technical Note TN 2353 on EPD database creation for building materials [42], which reports hard-coded extraction scripts written per-product-category. The authors explicitly identify the limits of this approach: the considerable time spent writing the scripts, the high specificity to a single product category, and applicability only to completely standardised EPD formats. The note further observes that even within a product category, there is often a lack of consistency in reporting between EPDs. This is the precise argument for the routing approach proposed here.

Olanrewaju, Enegbuma, Donn and Oyefusi's 2025 paper on EPD database quality assurance [43] adopted a mixed methodology including NLP analysis of sample EPDs across operators, concluding that the poor association between EPDs published by different programme operators creates significant data quality challenges. Their findings directly support the programme-operator-as-complexity-dimension stratification proposed in this project's methodology. A companion paper by the same group [44] reviews wider challenges in EPD development and utilisation for whole-building life cycle assessment.

Broyles, Gevaudan and Brown's 2023 dataset compilation [45] reports successful extraction of LCA midpoints from 39,213 ready-mix concrete EPDs using a custom Python script, but is limited to a single product category and reports no error analysis or generalisation beyond ready-mix concrete. Earlier work by Koyampambath, Adibi, Szablewski, Adibi and Sonnemann [46] used NLP and random forest regression to *predict* impact category values rather than extract them, achieving R-squared correlations of 0.81 for Global Warming Potential and lower for other indicators on EPDs sourced from the ÖKOBAUDAT platform.

To the best of this author's knowledge, no published work has applied modern LLM-based or SLM-based extraction methods to a heterogeneous EPD corpus across multiple programme operators with a principled evaluation across the cost-quality frontier. This is the gap the present project addresses.

3.4 Supporting Technical Literature

Three further bodies of work inform the project's methodology.

PDF and layout-aware extraction. The IBM Docling project [47] provides the current state-of-the-art open-source pipeline for document conversion, integrating the TableFormer transformer model [48] for table structure recognition (reporting tree-editing-distance scores of 98.5% on simple tables and 95% on complex tables). Docling will form the preprocessing layer of the proposed pipeline. Earlier work on layout-aware language models (LayoutLMv3 [49] and the VILA family [50]) established the importance of jointly modelling text and 2D position. The Colakoglu, Solmaz and Fürst LayIE-LLM paper [51], published at EMNLP Findings 2025, benchmarks the design space of layout-rich document extraction with LLMs and is the most directly comparable methodological reference for this project.

Structured extraction and constrained decoding. The XGrammar engine of Dong et al. [35], now the default structured-generation backend for vLLM and SGLang, achieves up to 100× speedup over earlier grammar-constrained approaches. The JSONSchemaBench benchmark [37] established a standardised evaluation methodology for schema-constrained extraction. The counter-argument that format restrictions can hurt accuracy [52] must also be engaged with.

Confidence calibration and uncertainty quantification. The cascade design proposed in Chapter 4 requires a confidence signal to decide whether a document handled by the SLM tier should be escalated. The relevant literature includes Wang et al.'s self-consistency sampling [53], Kuhn, Gal and Farquhar's semantic uncertainty work [54], Farquhar et al.'s Nature paper on hallucination detection via semantic entropy [55], and recent surveys of uncertainty quantification in LLMs [56]. These provide the methodological vocabulary for the project's evaluation of confidence calibration as both a cascade signal and a qualitative metric.

Chapter 4 — Project Plan

4.1 Methodology

The project methodology is organised around FrugalGPT's three frugal strategies. Each strategy is implemented as a component of the pipeline and evaluated independently and in combination.

4.1.1 Strategy 1: Prompt Adaptation

The Firstplanit schema requires approximately 210 fields per document, of which 168 form the structured indicator $\times$ stage matrix. Naive prompting that requests all fields in a single pass risks excessive prompt length and degraded response quality. This project evaluates two prompt-adaptation tactics: (i) schema chunking, in which the indicator $\times$ stage matrix is requested in sub-blocks (e.g. by life-cycle stage group) and recombined post-hoc, and (ii) guided decoding using XGrammar [35] via vLLM, which constrains output to schema-valid JSON at the token level and eliminates the need for verbose schema-instruction prefixes in the prompt itself. Prompt-adaptation effectiveness will be measured by tokens-per-document (input and output) and accuracy retention against an unconstrained baseline.

4.1.2 Strategy 2: LLM Approximation (Local Fine-Tuning)

The novel extension of this project lies primarily in this strategy. Where FrugalGPT's original "approximation" tactic substitutes a cheaper cloud API for a more expensive one, this project substitutes a *domain-fine-tuned local SLM*, specifically, an EPD-fine-tuned variant of Phi-4-Mini or Qwen 2.5 3B for both. Fine-tuning will use QLoRA [17] on a training subset of the gold-standard EPDs (or a programmatically augmented version thereof), following the Strata methodology [18] which demonstrates that fewer than 100 high-quality training examples can achieve human-level extraction on desktop GPUs.

Concretely, the planned fine-tuning protocol is: select Phi-4-Mini (3.8B) as the base model for its strong reasoning at small scale [14]; construct training pairs from the gold-standard set using stratified sampling across complexity dimensions; apply QLoRA with 4-bit quantisation; evaluate against the zero-shot and few-shot variants of the same base model. At least one fine-tuned model will be delivered; fine-tuning of mid-tier (7–14B) models is a stretch goal contingent on time and hardware availability.

4.1.3 Strategy 3: LLM Cascade

The cascade combines Strategies 1 and 2. The proposed cascade has three tiers: (i) the fine-tuned local SLM with guided decoding (Strategy 2 applied to Strategy 1); (ii) a larger local model — either an unmodified mid-tier model or, if Strategy 2 stretch is reached, a fine-tuned 7–14B model; (iii) a frontier cloud model (Claude / GPT / Gemini 2.5 Pro) as terminal fallback. Following FrugalGPT's original mechanism, cascade decisions will be made on the basis of *output confidence* from the SLM tier using a combination of token-level probability, self-consistency sampling [53], and semantic entropy [54] as candidate signals to be empirically compared.

The cascade is evaluated on the basis of: cost per document at chosen confidence thresholds, accuracy vs. always-frontier baseline, and the cost-accuracy Pareto frontier traced by sweeping the threshold.

4.1.4 Pipeline Implementation

The strategies above are implemented within a four-stage extraction pipeline:

1. **Preprocessing.** Native PDF inspection with tag detection, OCR fallback (Tesseract or PaddleOCR) for scanned documents, language detection, layout-aware table extraction via Docling [47].
2. **SLM tier extraction.** Fine-tuned local SLM with guided-decoding output, producing schema-valid JSON and a confidence signal.
3. **Cascade decision.** Confidence-based routing to (a) accept, or (b) escalate to next tier.
4. **Validation and output.** Schema validation, internal-consistency checks (e.g. the A1–A3 convenience field should reconcile with A1+A2+A3 where both are reported), and write-back to the Firstplanit MongoDB collection (or most likely a simulation of the startup's database).

4.1.5 Corpus and Gold Standard

The empirical foundation of the project is a hand-verified gold standard of 100 EPDs constructed from Firstplanit's production corpus, stratified across complexity dimensions identified in a preliminary corpus audit: PDF structure (tagged/untagged, native/scanned), document layout (table count, language), EPD content (single/multi-product, populated stages), programme operator, and issuance date. An additional ~10-EPD stress-test subset of edge cases will be constructed for failure-mode analysis. The EC3 / Building Transparency database [29] will serve as an independent cross-reference where overlap permits.

Discussions remain to be held with the industrial partner on access to this dataset. A contingency plan is to curate a new gold-standard set personally using open-sourced EPDs from *environdec.com*. The author's prior background in architecture allows for informed labelling under industry knowledge, helping ensure annotation quality.

4.1.6 Baselines and Evaluation Matrix

A central methodological feature of this project is the breadth of the evaluation matrix. The three FrugalGPT strategies are evaluated both *individually* and *in combination*, across multiple model sizes, against multiple baselines. This generates a large combinatorial space of conditions, which is itself part of the contribution.

The baselines and conditions to be evaluated are:

- **Baseline A — Firstplanit production:** the current Gemini 2.5 Flash pipeline as deployed by Firstplanit. The realistic comparison point.
- **Baseline B — Frontier cloud reference:** Claude Sonnet 4.7, GPT, or Gemini 2.5 Pro on the same prompt. The upper-bound reference.

- **Baseline C — Local SLM, not fine-tuned:** zero-shot Phi-4-Mini, Qwen 2.5 3B, Gemma 3 4B. Isolates the cost of running locally without domain adaptation.
- **Strategy 1 alone — Prompt adaptation:** applied to each of the baselines above. Measures the contribution of chunking and guided decoding.
- **Strategy 2 alone — Local fine-tuning:** fine-tuned Phi-4-Mini (and stretch: fine-tuned mid-tier) against its non-fine-tuned counterpart. Isolates the contribution of approximation.
- **Strategy 3 alone — Cascade:** two-tier and three-tier cascades using non-fine-tuned local SLMs as the entry point. Isolates the contribution of cascading without approximation.
- **Combined: S1 + S2:** fine-tuned SLM with prompt adaptation.
- **Combined: S1 + S3:** cascaded non-fine-tuned models with prompt adaptation.
- **Combined: S2 + S3:** cascade with fine-tuned entry tier.
- **Combined: S1 + S2 + S3 (full pipeline):** the proposed end-state, evaluated as the principal contribution.

Each condition is evaluated on the gold standard, with multiple confidence thresholds where applicable, producing a populated cost-quality Pareto frontier. Where time and budget permit, conditions are run across multiple model sizes within a tier (e.g. Phi-4-Mini 3.8B alongside Qwen 2.5 3B) to isolate model-architecture effects from strategy effects.

A practical consequence of this matrix is that not every cell can necessarily be executed within the project timeline. Section 4.3 discusses prioritisation: Baselines A and B and the full S1+S2+S3 pipeline are protected as must-haves; individual-strategy conditions are protected for Strategy 2 (fine-tuning), whose contribution most affects the headline claim; and additional model-size variations are stretch deliverables.

4.1.7 Evaluation Framework

Quantitative metrics, measured per strategy, cascade configuration, and baseline, comprise: per-field accuracy (exact-match for categorical, tolerance-based for numeric); precision, recall, and F1 *disaggregated by field category* (a critical methodological point, since approximately 88 of the 168 indicator × stage cells are validly N/A and a model returning N/A everywhere would score misleadingly high on naive accuracy); schema validity rate; input and output token counts; p50/p95 latency; cost per document (cloud API spend or amortised GPU-hour cost for local); and hardware footprint at chosen quantisation. Qualitative analysis comprises confidence calibration (does the SLM's stated confidence correspond to its actual accuracy?) and a systematic failure-mode taxonomy. Beyond this, the project will also seek to extend current debates on appropriate metrics beyond F1 and accuracy alone, to include cost, latency, and energy use across the lifecycle of the pipeline, as suggested by recent work where evaluation criteria must be reconsidered more carefully once Frugal principles extend beyond pure efficiency.

4.2 Success Criteria

Success will be evaluated against three concrete criteria:

- 1) **Empirical:** The project will deliver a reproducible comparative evaluation of the three FrugalGPT strategies, implemented individually and in combination, against Firstplanit's production Gemini 2.5 Flash baseline and a frontier cloud reference, on a complexity-stratified gold standard of 50–100 EPDs. The threshold of success is not that the routed pipeline beats Gemini 2.5 Flash on accuracy. It is that the project produces a defensible cost-quality Pareto frontier across the strategies tested and identifies at least one configuration that demonstrably improves on the baseline along at least one operationally meaningful axis (cost, latency, governance, or vendor independence) at an acceptable accuracy threshold. A value of 0.95 is commonly adopted in the studies reviewed, though the appropriate threshold for real-world deployment is itself an open question.
- 2) **Technical deliverable:** A working, reproducible pipeline implementing the three strategies on local hardware, with the SLM tier including at least one EPD-fine-tuned model. The pipeline is deployable within Firstplanit's infrastructure or a comparable startup environment. Code will be made available on GitHub or GitLab, with the possible extension of an application prototype to explore industry application considerations.
- 3) **Methodological:** A failure-mode taxonomy for EPD extraction, and a critical reflection on the deployment of frugal-AI strategies in a startup environment, including discussion of trade-offs between accuracy, cost, data governance, energy use across the entire lifecycle of the pipeline, and vendor independence that are not captured by accuracy-only evaluation. The reflection must also critically account for the additional resources consumed by routing and cascading themselves, ensuring that frugal claims are made net of implementation overhead.

4.3 Timeline

The remaining project schedule covers approximately 14 weeks from submission of this interim report to dissertation submission. The schedule allocates time according to the prioritisation logic discussed in Section 4.1.6: must-have conditions are run first, with stretch conditions reserved for later phases.

Period	Activities
Weeks 1–2	Complete corpus audit; finalise gold-standard construction; complete preliminary Docling pipeline runs
Weeks 3–4	Implement Baselines A, B, C; complete Strategy 1 (prompt adaptation) ablation runs
Weeks 5–7	Execute QLoRA fine-tuning of Phi-4-Mini (Strategy 2); evaluate fine-tuned vs non-fine-tuned local SLMs
Weeks 8–9	Implement cascade (Strategy 3); calibrate confidence thresholds; run two- and three-tier cascade conditions
Weeks 10–11	Combined-strategy evaluation (S1+S2+S3 full pipeline); failure-mode analysis; stretch model-size variations
Weeks 12–13	Industrial reflection chapter; pipeline documentation; final cost-quality Pareto-frontier visualisations
Week 14	Final write-up, revisions, and submission

Several risks could materially affect this timeline. *First*, a sufficient quantity and diversity of ground-truth data must be secured to ensure the validity of the experiments run; Firstplanit is understandably cautious about oversharing proprietary data, and therefore a portion of the gold standard may need to be manually labelled by the author, which is time-consuming and low value adding. *Second*, the corpus audit may surface complexity dimensions not yet anticipated, requiring re-stratification of the gold standard. *Third*, local hardware constraints may limit the size of fallback models testable on-premise, requiring more cloud-API usage than ideal. These are partly mitigated by the prioritisation logic in Section 4.1.6: the protected must-have conditions (Baselines A and B, Strategy 2 fine-tune, S1+S2+S3 combined) can be executed independently of stretch components, ensuring a complete primary evaluation regardless of secondary slippage. Finally, technical difficulties at any implementation stage could significantly delay the timeline; as implementation has not yet started, the most complex stage is not yet known, and contingency buffer may need to be redistributed in response.

4.4 Ethics

This project involves no human subjects, no personal data, and no clinical or otherwise sensitive material. The EPD corpus consists of commercially issued declarations on construction products, which are largely available in open form through programme operator websites.

Particular care must be taken with respect to proprietary information from the industrial partner. A non-disclosure agreement has been signed between the author and Firstplanit, recognising that this is an academic project that will be reviewed by the supervisor and a second marker who cannot themselves be parties to that NDA. As a consequence, the author must ensure that this report and the final dissertation do not disclose: the full set of data points Firstplanit extracts from EPDs, full technical details of the production extraction pipeline as currently deployed, or the entire data sets shared with the author by Firstplanit (should such sharing take place). Aggregated findings, comparative results, and methodological choices remain reportable; specific commercial details do not.

Finally, this project will be transparent and academically rigorous in its use of AI coding agents and writing assistants. These tools are used in the development workflow for efficiency, but all original intellectual contributions remain with the author, and all referenced material is properly cited. A summary of AI-tool usage will be included in the final dissertation as a methodological appendix.

No Imperial ethical review is required for this work. The Imperial ethics self-assessment has been completed and is appended as Appendix A.

4.5 Open Questions for Supervisor Discussion

Several scoping and methodological decisions remain genuinely open and will be discussed with the supervisor and industrial partner before finalisation. They are documented here both for transparency and to invite feedback from the second marker.

- 4) **Project framing.** The current framing positions Frugal AI as the high-level conceptual umbrella under which the project sits. An alternative would lead with the EPD extraction problem and

Firstplanit's commercial context, treating Frugal AI as one of several tools applied to that problem. The concern with the current framing is whether it risks detracting from the core thesis — building a better data extraction pipeline for the industrial partner — by abstracting it under a broader philosophical movement. The choice has implications for the opening of the dissertation and for how the contribution is positioned.

- 5) **Whether to incorporate upfront complexity-aware routing as an extension.** The methodology above uses confidence-based cascade decisions following FrugalGPT's original mechanism. An alternative — or complementary — design would use *upfront* document features (tagged/untagged, language, page count, programme operator) to route documents before the SLM tier is even invoked, potentially saving inference cost on documents that are predictably difficult. This was originally a central element of the project but has been demoted to a possible extension on the grounds that (a) it adds methodological complexity that risks diluting the core contribution and (b) FrugalGPT's confidence-based cascade is already a validated mechanism for the same problem, just with a different signal. A modest experiment comparing the two routing signals on a subset of the gold standard is a candidate stretch goal if time permits.
- 6) **Gold-standard size.** Hand-verification is the rate-limiting step. Fifty EPDs or fewer is achievable within the project timeline and provides adequate statistical power for the primary comparisons; one hundred would enable more granular per-stratum analysis but at significant verification cost. A pilot of 30 hand-verified EPDs is being used to estimate inter-annotator agreement and verification time-per-document before committing.
- 7) **Whether to include openEPD or ECO Platform machine-readable data as a separate baseline.** A growing minority of EPDs are available in both PDF and machine-readable openEPD format [30]. Where overlap exists, the machine-readable version effectively represents a "perfect extraction" reference. Whether to use this as an additional baseline or only as a sanity-check cross-reference is an open methodological question. A deeper version of the same question: machine-readable EPDs are still an emerging standard, but their gradual adoption raises the question of whether they could in time invalidate the motivation for PDF-based EPD extraction entirely, or whether the installed base of legacy declarations keeps the problem alive for the foreseeable horizon.
- 8) **Handling of complete extraction failures.** Some EPDs may resist all three cascade tiers — typically the very oldest scanned multi-product cases. The pipeline must choose between flagging for human review, accepting partial extraction, or raising a hard failure. The right policy depends on Firstplanit's downstream tolerance for missing-versus-incorrect data, which has not yet been determined. These edge cases also raise interesting questions on the industry-application dimension of frugal AI: how routing pipelines degrade gracefully under genuinely adversarial documents is a novel extension of current frugal AI literature, which tends to assume documents are extractable in principle.
- 9) **Confidence calibration evaluation methodology.** The confidence signal is central to the cascade. Evaluating *whether* a model's confidence is well-calibrated requires comparing

stated confidence to actual correctness, which in turn requires the gold standard. Whether a separate held-out calibration set is needed, or whether cascade-escalation patterns on the existing gold standard provide adequate signal, remains an open question.

10) Project scope. A more open-ended question on the project's overall scope and ambition. Is the current plan appropriately scoped for an MSc dissertation, assuming successful implementation? The current approach builds primarily on top of the FrugalGPT framework to provide a focused structure and academic validity, with the extension into fine-tuned local SLMs as the principal methodological novelty. The question for discussion is whether this represents sufficient research novelty for a project at this level, or whether the dissertation should aim to develop its own framework — for instance, a complexity-aware routing framework or a domain-specific extraction methodology — to satisfy the research-contribution expectations of an MSc.

Chapter 5 — Progress to Date

5.1 Activities and Accomplishments

Work to date has concentrated on four areas. *First*, an extensive literature review across the Frugal AI movement, FrugalGPT and its successors, SLM-LLM collaboration, layout-aware document understanding, structured extraction, confidence calibration, and analogous industrial extraction work in finance, science, and clinical informatics. The output of this review forms Chapter 3 of this report and will be carried directly into the final dissertation.

Second, scoping discussions with Firstplanit have confirmed the operational schema (approximately 210 fields per document, of which 168 form the structured indicator × stage matrix), access to the production MongoDB collection, and the current production baseline (Gemini 2.5 Flash). A working understanding of Firstplanit's ingestion architecture, the points at which the proposed pipeline would integrate, and overall business operations has been established.

Third, a preliminary technical survey of the deployment landscape has been conducted, identifying Docling as the preprocessing tool of choice for now, vLLM with XGrammar as the local inference and guided-decoding stack, the candidate SLM tier (Phi-4-Mini, Qwen 2.5 3B/7B, Gemma 4 4B/12B, Qwen2.5-VL) as well as relevant GitHub repositories to reference. A development environment has been established on Imperial computing resources and tested with single-pass Docling conversions on a small sample of EPDs.

Fourth, the schema, complexity-stratification approach, and evaluation matrix have been sketched-out and documented in this report.

5.2 Obstacles Encountered and Mitigations

Three obstacles have arisen and are being managed.

The first lack detail on current Firstplanit benchmarks. An internal report has been shared regarding the costs per document between different models however metrics measure time and cost only with no further breakdowns in terms of accuracy or time spent reviewing or energy. Mitigation: Re-deploy their current pipeline and conduct experiment to obtain more fine-grained benchmarks for comparison.

The second is the absence of published baselines or gold-standard datasets for EPD extraction. Unlike SEC filings or clinical notes, there is no public benchmark against which to evaluate. Mitigation: Manually gold-standard construct, with the construction will serve as both this project's evaluation set and a potential contribution to the public literature, conditional on permissions from Firstplanit.

The third is the cost and latency overhead of running multiple frontier cloud APIs as baselines and reference points within a student computing budget. The breadth of the evaluation matrix (Section 4.1.6) compounds this. Mitigation: API spend is being concentrated on the gold-standard subset, with rate-limited and quantised local runs used for development and iteration; the prioritisation logic in Section 4.1.6 ensures the most important conditions are protected.

5.3 Plans for Remaining Work

The immediate next steps, covering the four weeks following submission of this report, are to complete the corpus audit, finalise the gold-standard set, and produce a first end-to-end execution of Baseline C (non-fine-tuned local SLM) on a representative sample. By the midpoint of the remaining project window the fine-tuned Phi-4-Mini will be trained and benchmarked, and the cascade will be implemented and calibrated. The final weeks are reserved for combined-strategy evaluation, failure-mode analysis, industrial reflection, and write-up. The stretch goals (upfront complexity-aware routing, mid-tier fine-tuning, and additional model-size variations) will be activated only once the protected primary evaluation is secure, and if deemed relevant from conversations to follow with the supervisor, second marker, and Firstplanit.

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